Yiming Liu

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Education

Sun Yat-sen University, Guangzhou 2022 – 2025
M.Eng. in Computer Vision
HCP Lab, advised by Prof. Liang Lin *(IEEE Fellow)*

Dalian University of Technology, Dalian 2018 – 2022
B.Eng. in Electronic and Information Engineering GPA : 4.59/5, Rank 4/206

Experience & Research

LLM Evaluation (Tencent Hunyuan) Jul 2025 – Present

- **Benchmarks**: built **VisualNeedle** (active visual search in information-dense scenes, whose counterfactual crop-black setting verifies a model really uses the evidence tools return), **jy-crpg-bench** (long-horizon play in an open-world RPG) and **EternalMath** (a math benchmark that grows with human discovery)
- **Video scoring model**: split video quality into text–image consistency, visual quality and dynamic quality, built a 10k in-house dataset, and compared SFT, RL and frozen-scoring recipes on VLMs, reaching SOTA on in-house and public benchmarks (**Q-Save**)

OpenRLHF: Alignment Algorithms & Training Framework Sep 2024 – Present

- **Framework**: led the KL-loss algorithms in OpenRLHF and built training backends including DeepSpeed AutoTP and FSDP2+CP+TP
- **Algorithm**: first identified that the `k3` estimator is biased when differentiated as a loss in GRPO, and proposed the unbiased `k2` loss; cited by Reinforce++ and expanded into **Rethinking KL Regularization in RLHF**

Diffusion-Based Adversarial Purification (HCP Lab, SYSU) Mar 2023 – Apr 2025

- **Analysis**: argued that diffusion purification defends through stochasticity rather than through forward noising shrinking the clean–adversarial KL divergence or a masking effect of Gaussian noise, and that its local loss landscape is not flat; first to separate this stochasticity from the one behind certified robustness
- **Method**: proposed diffusion denoising adversarial training, which smooths the loss landscape and improves robustness consistently across DDPM, DDIM, VP-SDE and EDM

MMDetection Framework Development (Shanghai AI Lab) Jul 2022 – Mar 2023

Core contributor to the Detection Transformer refactor across MMDetection 2.0 and 3.0

- Simplified the registry and inheritance hierarchy and unified module data formats, improving readability, maintainability and runtime efficiency; MMDetection opened the branch refactor-detr for this work
- Re-implemented Conditional/DAB DETR and DINO, settled their naming and abstractions, aligned the reproduced metrics strictly and documented them; presented at an OpenMMLab community talk

Publications

- **ICLR 2025** *(first author)* Towards Understanding the Robustness of Diffusion-Based Purification: A Stochastic Perspective
- **EMNLP 2025 Demo** OpenRLHF: A Ray-based Easy-to-use, Scalable and High-performance RLHF Framework
- **ICASSP 2025** Text Enhancement Network for Complex Multi-line Scene Text Image Super-resolution
- **Under review** *(second author)* jy-crpg-bench: A Long-Horizon Benchmark for Frontier Models in an Open-World RPG
- **Under review** *(co-first author)* VisualNeedle: Benchmarking Active Visual Search in Information-Dense Scenes
- **arXiv** Rethinking KL Regularization in RLHF: From Value Estimation to Gradient Optimization
- **arXiv** Q-Save: Towards Scoring and Attribution for Generated Video Evaluation
- **arXiv** EternalMath: A Living Benchmark of Frontier Mathematics that Evolves with Human Discovery
- **arXiv** DramaChain Bench: An End-to-End Benchmark for Short-Drama Generation

Skills

- Interests: LLM evaluation (benchmark design, automatic scoring and attribution), Agentic RL
- Training: DTensor-based FSDP2 parallelism, DeepSpeed, Ray + vLLM RLHF stack